Perimeters and areas of similar figures

Core Concepts: Perimeters and areas of similar figures

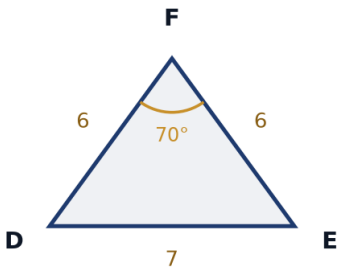
1 In $\triangle ABC$ studying perimeters and areas of similar figures, $m\angle A = 40^\circ$ and $m\angle B = 45^\circ$. What is $m\angle C$?

a. 95°

b. 105°

c. 87°

d. 140°



2 A geometric figure scaled by factor $k = 2$ has an original side length of 4 cm. What is the corresponding image length?

a. 6 cm

b. 8 cm

c. 12 cm

d. 16 cm

3 On the coordinate plane, a line with slope $m = 5$ passes through $(0, 3)$. What is its y-coordinate when $x = 1$?

a. 3

b. 10

c. 8

d. 13

Procedural Fluency & Algebraic Manipulation

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4 In $\triangle ABC$ studying perimeters and areas of similar figures, $m\angle A = 40^\circ$ and $m\angle B = 52^\circ$. What is $m\angle C$?

a. 88°

b. 98°

c. 80°

d. 140°

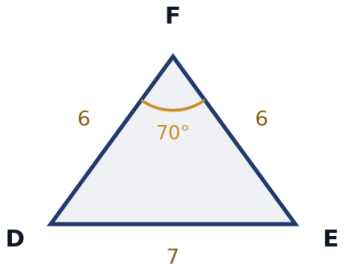
5 A geometric figure scaled by factor $k = 3$ has an original side length of 4 cm. What is the corresponding image length?

a. 9 cm

b. 12 cm

c. 16 cm

d. 36 cm



6 On the coordinate plane, a line with slope $m = 5$ passes through $(0, 3)$. What is its y-coordinate when $x = 2$?

a. 8

b. 15

c. 13

d. 18

Graphical & Geometric Interpretation

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7 In $\triangle ABC$ studying perimeters and areas of similar figures, $m\angle A = 40^\circ$ and $m\angle B = 59^\circ$. What is $m\angle C$?

a. 81°

b. 91°

c. 73°

d. 140°

8 A geometric figure scaled by factor $k = 4$ has an original side length of 4 cm. What is the corresponding image length?

a. 12 cm

b. 16 cm

c. 20 cm

d. 64 cm

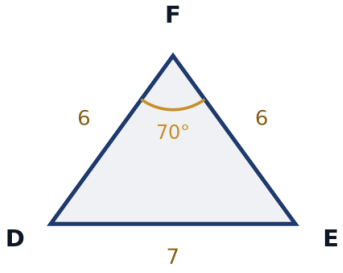
9 On the coordinate plane, a line with slope $m = 5$ passes through $(0, 3)$. What is its y-coordinate when $x = 3$?

a. 13

b. 20

c. 18

d. 23



Exam-Style Problem Solving & Modeling

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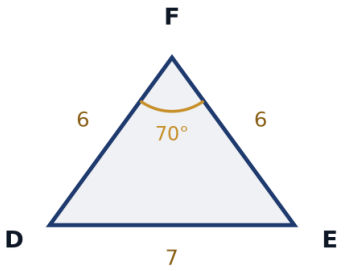
10 In $\triangle ABC$ studying perimeters and areas of similar figures, $m\angle A = 40^\circ$ and $m\angle B = 66^\circ$. What is $m\angle C$?

a. 74°

b. 84°

c. 66°

d. 140°



11 A geometric figure scaled by factor $k = 5$ has an original side length of 4 cm. What is the corresponding image length?

a. 15 cm

b. 20 cm

c. 24 cm

d. 100 cm

12 On the coordinate plane, a line with slope $m = 5$ passes through $(0, 3)$. What is its y-coordinate when $x = 4$?

a. 18

b. 25

c. 23

d. 28

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